

1) Given $E \rightarrow E + E / id$

$E \rightarrow E + E / id$

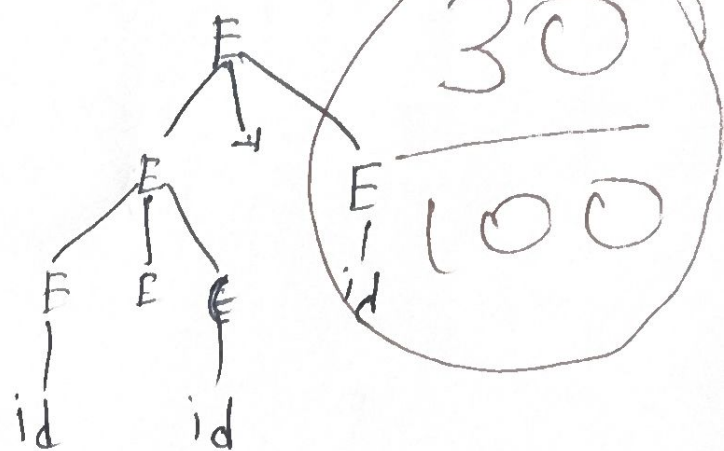
$id + id + id$

$E \rightarrow * E$

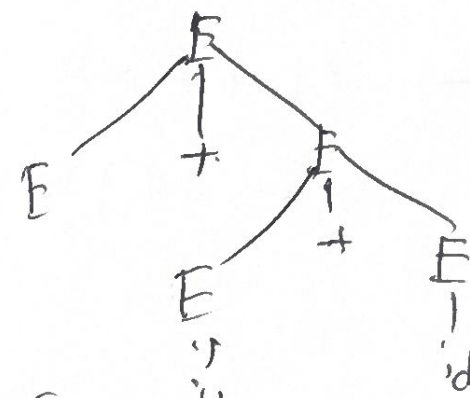
$E \rightarrow E * E'$

$E \rightarrow \cdot E / \epsilon$

E



$(id + id) + id - (1)$

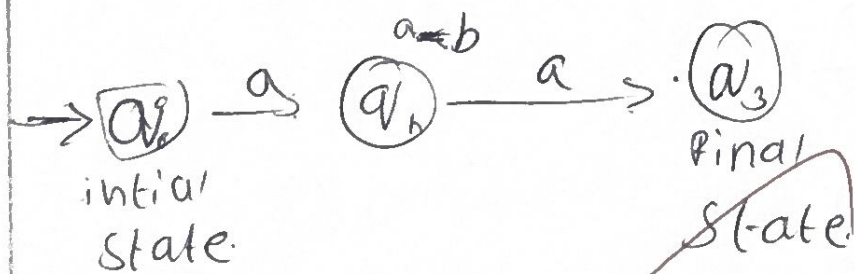


$id + (id + id) - (2)$

3) (1) = (2)

Ambiguous

2) $a \cdot (a/b)^* a$



Φ

Accepting strings

$a a b a a b a$

$a a b b a a$

$a b a$

$a a a$

$a b b a$

$$S \rightarrow iEts | iEts'$$

$$S \rightarrow iE + S S'$$

$$S' \rightarrow es / \epsilon$$

i) left factoring

$$S \rightarrow iE + S$$

$$S \rightarrow iE + S S$$

$$S \rightarrow a$$

$$S \rightarrow iE + S S'$$

$$S \rightarrow S \cdot es / \epsilon$$



$$S \rightarrow AB D h$$

$$B \rightarrow \cdot C ($$

$$C \rightarrow | C / \epsilon$$

$$D \rightarrow \cdot \epsilon F$$

$$F \rightarrow S g / \epsilon$$

$$F \rightarrow F / \epsilon$$



7) $S \rightarrow Sa/Sb/cd$

$A \rightarrow A\epsilon$

$S \rightarrow Sa$

$S \rightarrow Sb$

$S \rightarrow cd$

$S \rightarrow$

The given equation

8)

$E \rightarrow + E'$

$E' \rightarrow +Tt'/\epsilon$

$\text{First}(E') = \{+, \epsilon\}$

$$S \rightarrow \cdot (() S / \epsilon$$

$$S \rightarrow ((\epsilon) \cdot S$$

$$((((S)) S) S$$

$$(((\epsilon) (\epsilon) \epsilon$$

$$(())$$

∴ It is a valid string

Given

$$S' \rightarrow SS$$

$$S \rightarrow \cdot (($$

$$(\rightarrow S ((d$$

$$\text{item: } (S' \rightarrow \cdot S)$$

$$S' \rightarrow SS$$

$$\underline{I_0} : S' \rightarrow S S'$$

$$S \rightarrow (($$

$$(\rightarrow \cdot (($$

$$(\rightarrow d$$

12) $S \rightarrow AB$ $A \rightarrow a$ $B \rightarrow b$

$\text{First}[S] = \{A, B\}$

$\text{First}(A) = \{a\}$

$\text{First}(B) = \{b\}$

$S \rightarrow a \cdot b / a \cdot b$

$S \rightarrow a \cdot b$

$S \rightarrow A$



It accept aabb

✓

14) `#include <stdio.h>`

`int main () {`

`int a, b, sum;`

`a = 10;`

`b = 5;`

`sum = a + b;`

`printf ("%d", sum);`

`}`

`return 0;`

`}`

integer a

integer b

expression = $sum = a + b$

operator = $+$

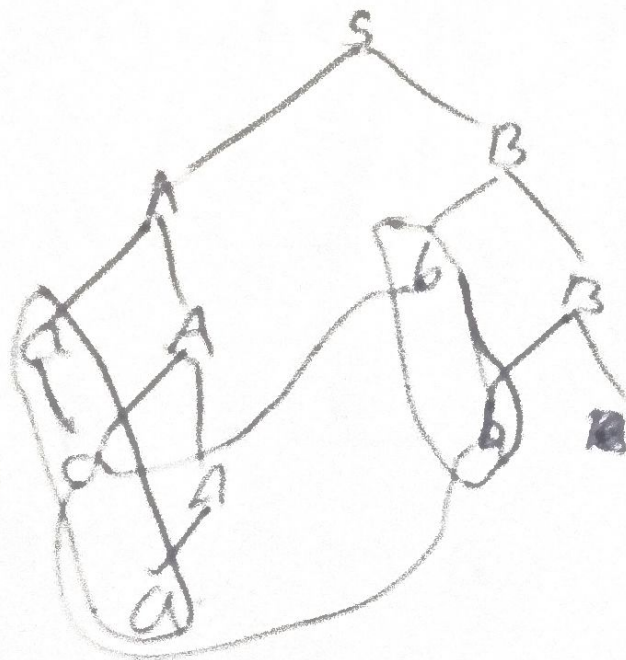
3

15)

$S \rightarrow AB$

$A \rightarrow aA/a$

$B \rightarrow bB/b$



$$\begin{aligned}
 (6) \quad & S \rightarrow SAB \\
 & A \rightarrow aA/\epsilon \\
 & B \rightarrow bB/\epsilon
 \end{aligned}$$

?

17) Left-Recursion.

Given.

$$A \rightarrow A\alpha/\beta$$

$$A \rightarrow \gamma A$$

$$A' \rightarrow \delta A'/\epsilon$$

Left Factorisation

$$A \rightarrow \gamma \alpha \beta / \alpha \beta$$

$$A \rightarrow \gamma \delta A'$$

$$A' \rightarrow \delta / \epsilon$$

✓

8) Grammar

$E \rightarrow S \cdot T E'$

$E' \rightarrow + T E' / \epsilon$

$T \rightarrow S \cdot F T'$

$T' \rightarrow S * F T' / \epsilon$

$F \rightarrow S \cdot (E) / id$

$First[E] = \{ \epsilon, + (T) \}$

$First[E'] =$



2

20) #include <stdio.h>

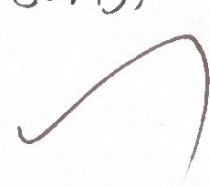
int main() {

int a = 10, b = 20;

int sum = a + b;

printf("%d", sum);

return 0;



}

a) Lexical Analyse

The code converted

into

tokens

id a, b, .

Syntax Analyzer

The code converted into Tree structure

$a + b$



Semantic Analyzer

It checks the code whether there is an error or not checks arrays...

Intermediate code

It converts source code into Assembly language

Code

Code optimizer

It optimizes the code it can minimize code.

It can help minimising the size

Code generator

This is the last phase It generates the

binary code. (0,1) The machine can understand

this code.

5)

